Lab 17.2 Seq2Seq Translation using Hugging Face

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## Install and Import Libraries

```
!pip install transformers sentencepiece -q

from transformers import pipeline, AutoTokenizer,
AutoModelForSeq2SeqLM
import pandas as pd
```

## Load Translation Models

```
lang="""STEP 2: Load Translation Pipelines
"""

# Custom Translator class to bypass pipeline issues
class CustomTranslator:
    def __init__(self, model_name):
        self.tokenizer = AutoTokenizer.from_pretrained(model_name)
        self.model = AutoModelForSeq2SeqLM.from_pretrained(model_name)

    def __call__(self, text):
        inputs = self.tokenizer(text, return_tensors="pt")
        outputs = self.model.generate(**inputs)
        # Mimic pipeline output format: a list of dictionaries
        return [{"translation_text": self.tokenizer.decode(outputs[0],
skip_special_tokens=True)}]

# English → French
translator_fr = CustomTranslator("Helsinki-NLP/opus-mt-en-fr")

# English → German
translator_de = CustomTranslator("Helsinki-NLP/opus-mt-en-de")

# English → Hindi
translator_hi = CustomTranslator("Helsinki-NLP/opus-mt-en-hi")

print("All models loaded successfully")
```

```
/usr/local/lib/python3.12/dist-packages/transformers/models/marian/
tokenization_marian.py:176: UserWarning: Recommended: pip install
sacremoses.
  warnings.warn("Recommended: pip install sacremoses.")
```

{"model_id":"1e632eb4a2f54edca8a66b61b0b46cff","version_major":2,"vers
ion_minor":0}

```
The tied weights mapping and config for this model specifies to tie
model.shared.weight to model.decoder.embed_tokens.weight, but both are
present in the checkpoints, so we will NOT tie them. You should update
the config with `tie_word_embeddings=False` to silence this warning
The tied weights mapping and config for this model specifies to tie
model.shared.weight to model.encoder.embed_tokens.weight, but both are
present in the checkpoints, so we will NOT tie them. You should update
the config with `tie_word_embeddings=False` to silence this warning
```

{"model_id":"3147d52312ab4aa49c1b430c90402dc0","version_major":2,"version_minor":0}

```
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present in the checkpoints, so we will NOT tie them. You should update
the config with `tie_word_embeddings=False` to silence this warning
```

{"model_id":"8a205afec8784fee90adc82ea9c88865","version_major":2,"version_minor":0}

```
The tied weights mapping and config for this model specifies to tie
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model.shared.weight to model.encoder.embed_tokens.weight, but both are
present in the checkpoints, so we will NOT tie them. You should update
the config with `tie_word_embeddings=False` to silence this warning
```

```
All models loaded successfully
```

## Prepare Dataset

```python
"""
STEP 3: Prepare Input Sentences
Different types of sentences
"""

sentences = [
    "The weather is very pleasant today.",
    "I am going to the market to buy groceries.",
    "Artificial intelligence is transforming the world rapidly.",
    "Can you help me solve this problem?",
    "She completed her project before the deadline.",
    "India is known for its diverse culture and traditions.",
    "The train arrived late due to heavy rain.",
```

```
    "Machine learning models require large amounts of data.",
    "He was tired, but he continued working.",
    "What are your plans for the weekend?",
    "The professor explained the concept very clearly.",
    "Social media has both positive and negative effects.",
    "We should reduce plastic usage to save the environment.",
    "The software system failed due to a critical error.",
    "Learning new skills is important for career growth."
]

len(sentences)
```

```
15
```

## Perform Translation

```
"""
STEP 4: Translate Sentences
"""

results = []

for sentence in sentences:

    fr = translator_fr(sentence)[0]['translation_text']
    de = translator_de(sentence)[0]['translation_text']
    hi = translator_hi(sentence)[0]['translation_text']

    results.append({
        "English": sentence,
        "French": fr,
        "German": de,
        "Hindi": hi
    })

    print("\nOriginal:", sentence)
    print("French :", fr)
    print("German :", de)
    print("Hindi  :", hi)
```

```
Original: The weather is very pleasant today.
French : Le temps est très agréable aujourd'hui.
German : Das Wetter ist heute sehr angenehm.
Hindi  : मौसम आज बहुत सुखद है.

Original: I am going to the market to buy groceries.
French : Je vais au marché pour acheter des courses.
German : Ich gehe auf den Markt, um Lebensmittel zu kaufen.
Hindi  : मैं बाजार के लिए किचक खरीदने के लिए जा रहा हूँ.
```

```
Original: Artificial intelligence is transforming the world rapidly.
French : L'intelligence artificielle transforme rapidement le monde.
German : Künstliche Intelligenz transformiert die Welt schnell.
Hindi  : कलापूर्ण बुद्धि दुनिया को तेज़ी से बदल रही है।

Original: Can you help me solve this problem?
French : Pouvez-vous m'aider à résoudre ce problème ?
German : Können Sie mir helfen, dieses Problem zu lösen?
Hindi  : आप इस समस्या को हल करने में मेरी मदद कर सकते हैं?

Original: She completed her project before the deadline.
French : Elle a terminé son projet avant la date limite.
German : Sie hat ihr Projekt vor Ablauf der Frist abgeschlossen.
Hindi  : उसने अपनी परियोजना को मृतलाइन से पहले पूरा किया.

Original: India is known for its diverse culture and traditions.
French : L'Inde est connue pour sa culture et ses traditions diverses.
German : Indien ist für seine vielfältige Kultur und Traditionen
bekannt.
Hindi  : भारत अपनी विभिन्न संस्कृति और परंपराओं के लिए जाना जाता है ।

Original: The train arrived late due to heavy rain.
French : Le train est arrivé tard en raison d'une forte pluie.
German : Der Zug kam wegen heftigem Regen spät an.
Hindi  : ट्रेन भारी वर्षा के कारण कुछ समय बाद आ गयी ।

Original: Machine learning models require large amounts of data.
French : Les modèles d'apprentissage automatique nécessitent de
grandes quantités de données.
German : Maschinelle Lernmodelle erfordern große Datenmengen.
Hindi  : मशीन सीखने वाले मॉडलों के लिए बहुत मात्रा की आवश्यकता होती है।

Original: He was tired, but he continued working.
French : Il était fatigué, mais il a continué à travailler.
German : Er war müde, aber er arbeitete weiter.
Hindi  : वह थक गया था, लेकिन वह काम जारी रहा ।

Original: What are your plans for the weekend?
French : Quels sont vos projets pour le week-end ?
German : Was haben Sie für ein Wochenende vor?
Hindi  : आप सप्ताहांत के लिए क्या योजना बना रहे हैं?

Original: The professor explained the concept very clearly.
French : Le professeur a expliqué très clairement le concept.
German : Der Professor erklärte das Konzept sehr deutlich.
Hindi  : प्रोफेसर ने इस धारणा को स्पष्ट रूप से समझाया ।

Original: Social media has both positive and negative effects.
```

```
French : Les médias sociaux ont des effets à la fois positifs et
négatifs.
German : Soziale Medien haben sowohl positive als auch negative
Auswirkungen.
Hindi  : सामाजिक मीडिया में भी अच्छे और बुरे अंजाम होते हैं।

Original: We should reduce plastic usage to save the environment.
French : Nous devrions réduire l'utilisation du plastique pour sauver
l'environnement.
German : Wir sollten den Einsatz von Kunststoff reduzieren, um die
Umwelt zu retten.
Hindi  : हमें पर्यावरण को बचाने के लिए प्लास्टिक का उपयोग कम करना चाहिए.

Original: The software system failed due to a critical error.
French : Le système logiciel a échoué en raison d'une erreur critique.
German : Das Softwaresystem ist aufgrund eines kritischen Fehlers
fehlgeschlagen.
Hindi  : गंभीर त्रुटि के कारण सॉफ्टवेयर तंत्र असफल.

Original: Learning new skills is important for career growth.
French : L'apprentissage de nouvelles compétences est important pour
la croissance de carrière.
German : Das Erlernen neuer Kompetenzen ist für das Karrierewachstum
wichtig.
Hindi  : नए हुनर सीखना, करियर बढ़ाने के लिए बहुत ज़रूरी है।
```

## Store Results in Table

```
"""
STEP 4: Display Results in DataFrame
"""

df = pd.DataFrame(results)
df
```


```
{"summary":"{\n  \"name\": \"df\",\n  \"rows\": 15,\n  \"fields\": [\n
{\n      \"column\": \"English\",\n      \"properties\": {\n
\"dtype\": \"string\",\n        \"num_unique_values\": 15,\n
\"samples\": [\n          \"What are your plans for the weekend?\",\n
\"Social media has both positive and negative effects.\",\n
\"The weather is very pleasant today.\"\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    },\n    {\n      \"column\": \"French\",\n      \"properties\":
{\n        \"dtype\": \"string\",\n        \"num_unique_values\": 15,\
n        \"samples\": [\n          \"Quels sont vos projets pour le
week-end ?\",\n          \"Les m\\u00e9dias sociaux ont des effets \\
u00e0 la fois positifs et n\\u00e9gatifs.\",\n          \"Le temps est
tr\\u00e8s agr\\u00e9able aujourd'hui.\"\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    },\n    {\n      \"column\": \"German\",\n      \"properties\":
```

```
{\n        \"dtype\": \"string\",\n        \"num_unique_values\": 15,\
n        \"samples\": [\n          \"Was haben Sie f\\u00fcr ein
Wochenende vor?\",\n          \"Soziale Medien haben sowohl positive
als auch negative Auswirkungen.\",\n          \"Das Wetter ist heute
sehr angenehm.\"\n        ],\n        \"semantic_type\": \"\",\n
\"description\": \"\"\n      }\n    },\n    {\n      \"column\":
\"Hindi\",\n      \"properties\": {\n        \"dtype\": \"string\",\n
\"num_unique_values\": 15,\n        \"samples\": [\n          \"\\
u0906\\u092a \\u0938\\u092a\\u094d\\u0924\\u093e\\u0939\\u093e\\
u0902\\u0924 \\u0915\\u0947 \\u0932\\u093f\\u090f \\u0915\\u094d\\
u092f\\u093e \\u092f\\u094b\\u091c\\u0928\\u093e \\u092c\\u0928\\u093e
\\u0930\\u0939\\u0947 \\u0939\\u0948\\u0902?\",\n          \"\\u0938\\
u093e\\u092e\\u093e\\u091c\\u093f\\u0915 \\u092e\\u0940\\u0921\\
u093f\\u092f\\u093e \\u092e\\u0947\\u0902 \\u092d\\u0940 \\u0905\\
u091a\\u094d\\u091b\\u0947 \\u0914\\u0930 \\u092c\\u0941\\u0930\\u0947
\\u0905\\u0902\\u091c\\u093e\\u092e \\u0939\\u094b\\u0924\\u0947 \\
u0939\\u0948\\u0902 \\u0964\",\n          \"\\u092e\\u094c\\u0938\\
u092e \\u0906\\u091c \\u092c\\u0939\\u0941\\u0924 \\u0938\\u0941\\
u0916\\u0926 \\u0939\\u0948.\"\n        ],\n        \"semantic_type\":
\"\",\n        \"description\": \"\"\n      }\n    }\n  ]\
n}","type":"dataframe","variable_name":"df"}
```


## Test Complex Sentences

```python
"""
STEP 7: Test Complex Sentences
"""

complex_sentences = [
    "Although the weather was bad, we decided to continue our journey.",
    "The system crashed because the server could not handle multiple requests.",
    "If you study regularly, you will achieve good results.",
    "He went to the office, finished his work, and returned home late.",
    "Why did you not inform me about the meeting earlier?"
]

for sentence in complex_sentences:

    fr = translator_fr(sentence)[0]['translation_text']

    print("\nOriginal:", sentence)
    print("French :", fr)
```

```
Original: Although the weather was bad, we decided to continue our
journey.
French : Bien que le temps ait été mauvais, nous avons décidé de
```

```
poursuivre notre voyage.

Original: The system crashed because the server could not handle
multiple requests.
French : Le système s'est écrasé parce que le serveur ne pouvait pas
gérer plusieurs requêtes.

Original: If you study regularly, you will achieve good results.
French : Si vous étudiez régulièrement, vous obtiendrez de bons
résultats.

Original: He went to the office, finished his work, and returned home
late.
French : Il s'est rendu au bureau, a terminé son travail et est rentré
tard chez lui.

Original: Why did you not inform me about the meeting earlier?
French : Pourquoi ne m'avez-vous pas informé de la réunion plus tôt ?
```

## Basic Analysis Output

```python
"""
STEP 6: Basic Analysis
"""

print("""
Observations:
- Translations are generally accurate for simple sentences.
- Complex sentences sometimes show slight grammatical changes.
- Technical terms are mostly preserved correctly.
- Hindi translations may slightly differ in sentence structure.
- Transformer models handle context better than traditional methods.
""")
```

```
Observations:
- Translations are generally accurate for simple sentences.
- Complex sentences sometimes show slight grammatical changes.
- Technical terms are mostly preserved correctly.
- Hindi translations may slightly differ in sentence structure.
- Transformer models handle context better than traditional methods.
```

## Conclusion

This assignment uses pretrained sequence-to-sequence transformer models to perform machine translation across multiple languages using Hugging Face pipelines.